

Final Lab Report

Faculty Initial : ABAH

Faculty Name : Mr. Abid Ahmad

Course Code and Section : CSE 472.3

Course Title : Web and Internet Programming Lab

Program : B.Sc in CSE

Department : Computer Science and Engineering

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Submitted By

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1. Objectives & Scope of the Application

Objectives:

- ❖ To develop an online crypto exchange application using HTML,CSS,JavaScript,PHP & MySQL
- ❖ To include two types of users : Admin & Trader/User each with different access privileges.
- ❖ To allow Users to:
 - Create an account, log in securely, and manage their crypto wallet.
 - View their assets, balance, and perform actions such as deposit, withdraw, trade, and transfer.
 - Check transaction history, rewards, and referral details.
- ❖ To allow Admins to:
 - Manage users, view transactions, and monitor trading activity.
 - Update crypto asset data, prices, and system configurations.
 - Oversee the database for performance and data consistency.
- ❖ To maintain a centralized MySQL database that stores user details, assets, orders, rewards, referrals, and settings.

Scope:

❖ Frontend:

Developed with HTML, CSS, and JavaScript to provide a responsive, dark-themed, and modern user interface for the exchange dashboard.

❖ User Module:

- User registration and secure login system.
- Dashboard displaying current crypto balance and asset overview.
- Perform trading actions like deposit, withdraw, and transfer.
- Access order history, rewards, and referral information.

❖ Admin Module:

- View and manage all users' data and transactions.
- Modify or remove crypto asset listings and price values.
- Handle system settings, referrals, and reward configurations.

❖ Database Integration:

Implemented using PHP for backend logic and MySQL for data storage.
All user information, wallet balances, and transaction records are stored securely.

❖ Security:

User authentication handled with PHP sessions and hashed passwords.
Prepared statements used for preventing SQL injection and data breaches.

❖ **Extensibility:**

The system can be expanded to include API-based real-time price updates, two-factor authentication, and live trading integrations in the future.

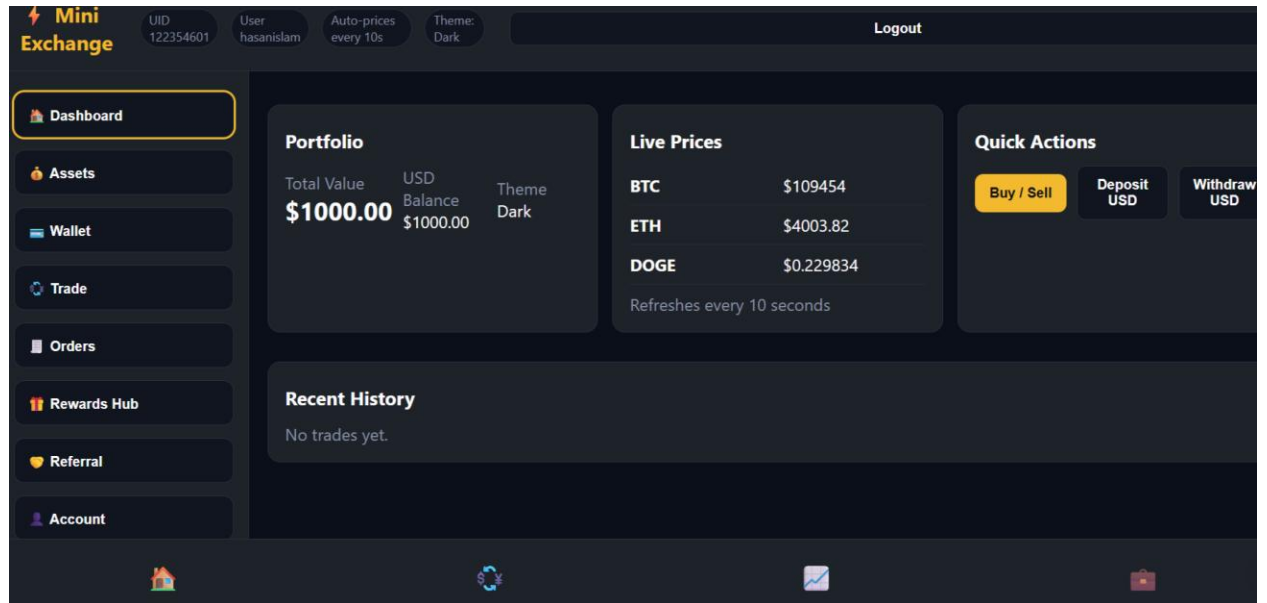
2. Features of the Website

- **Login & Registration Page :** The application begins with a secure login and registration page. Users must log in before accessing their dashboard. The admin has a separate panel with extended privileges.

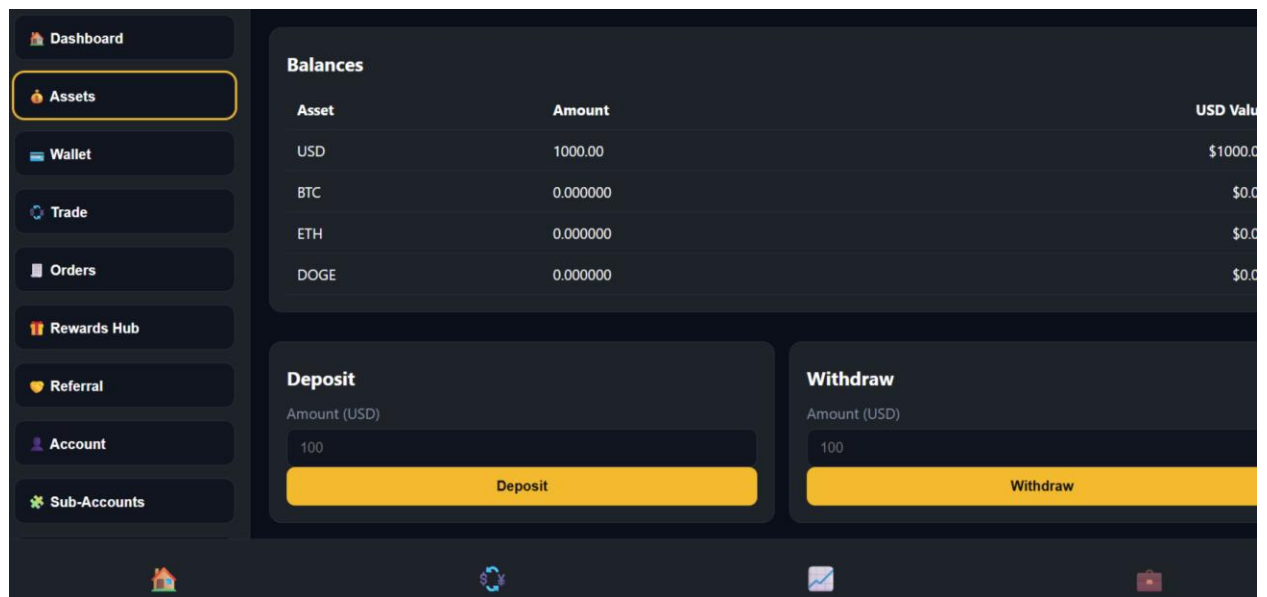
The image displays two side-by-side web forms for user authentication, set against a dark grey background with yellow accents. The left form is for registration, featuring fields for 'Username' (containing 'mdhasan'), 'Password', and 'Confirm Password' (all masked with dots). It includes a yellow 'Login' button and a yellow 'Sign up' button at the top, a yellow 'Create account' button below the password fields, and a grey 'Back to login' button at the bottom. The right form is for login, featuring fields for 'Username' (containing 'mdhasan') and 'Password' (masked with dots). It includes a yellow 'Login' button and a yellow 'Sign up' button at the top, a yellow 'Login' button below the password field, and a grey 'Quick demo' button at the bottom.

- **Navigation & Sections:**

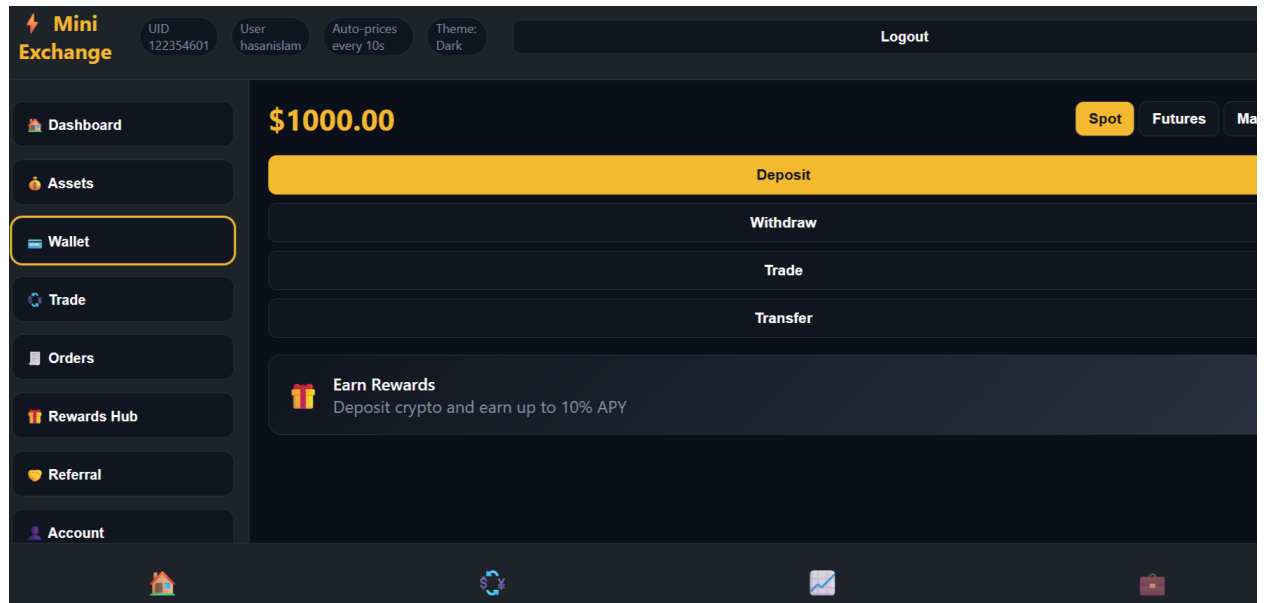
1. **Dashboard:** After login, users are redirected to their dashboard displaying wallet balance, assets, and recent activities.



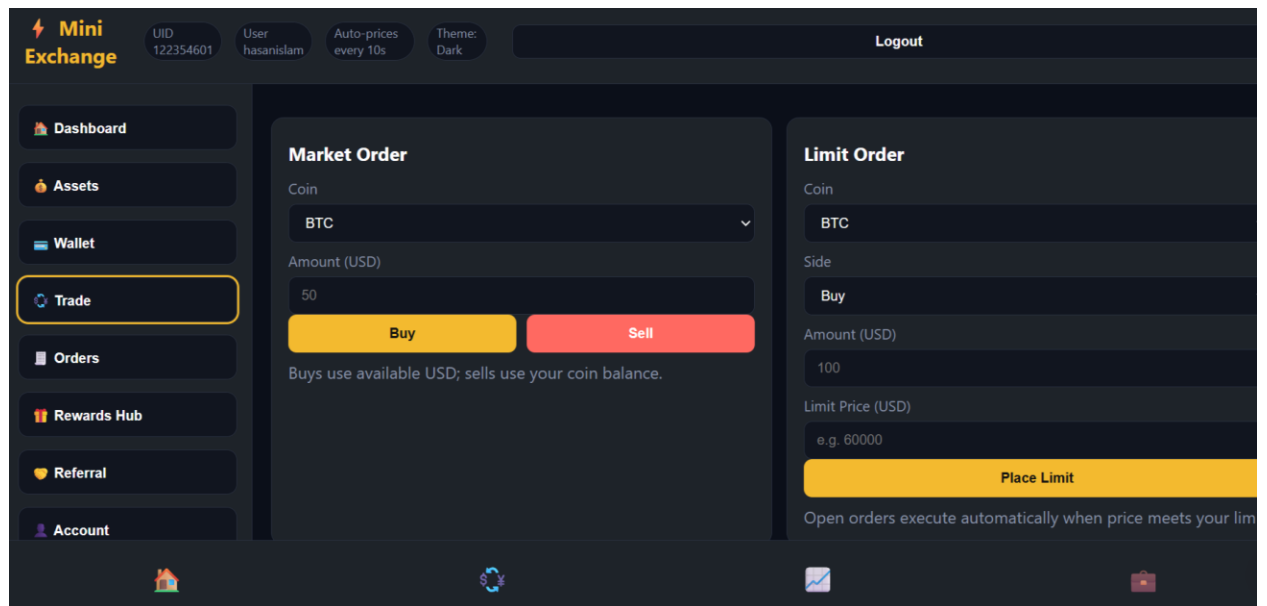
2. **Assets:** Displays available assets.



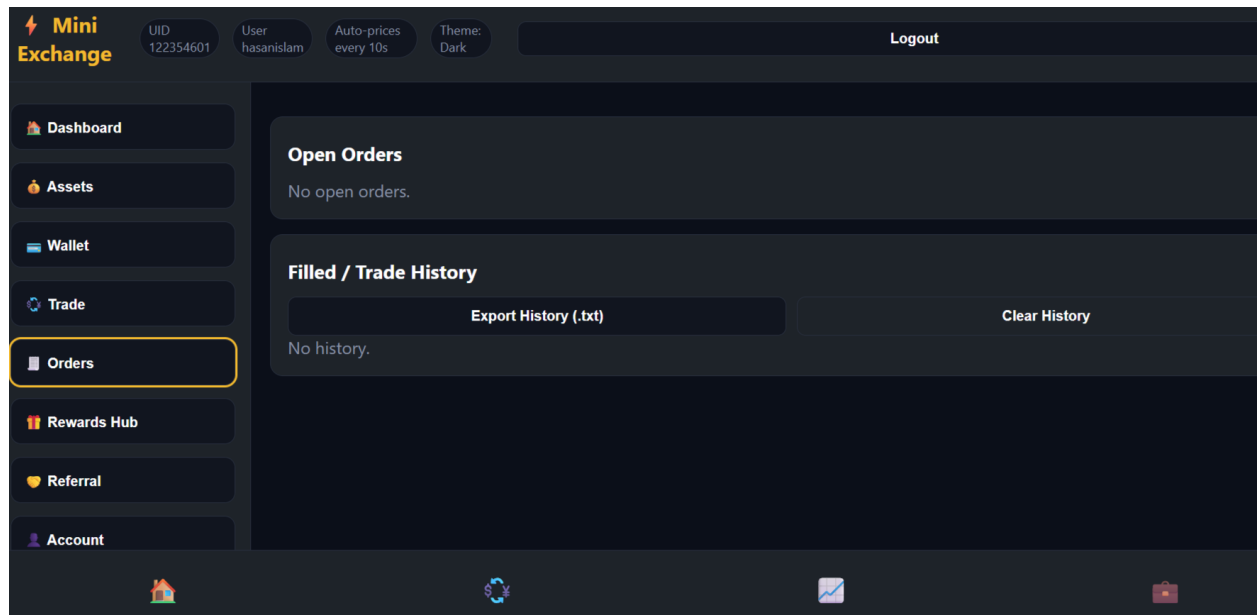
3. Wallet: Users can perform real-time **Deposit**, **Withdraw**, **Trade**, and **Transfer** actions. All operations are logged in the database and reflected instantly in their wallet balance.



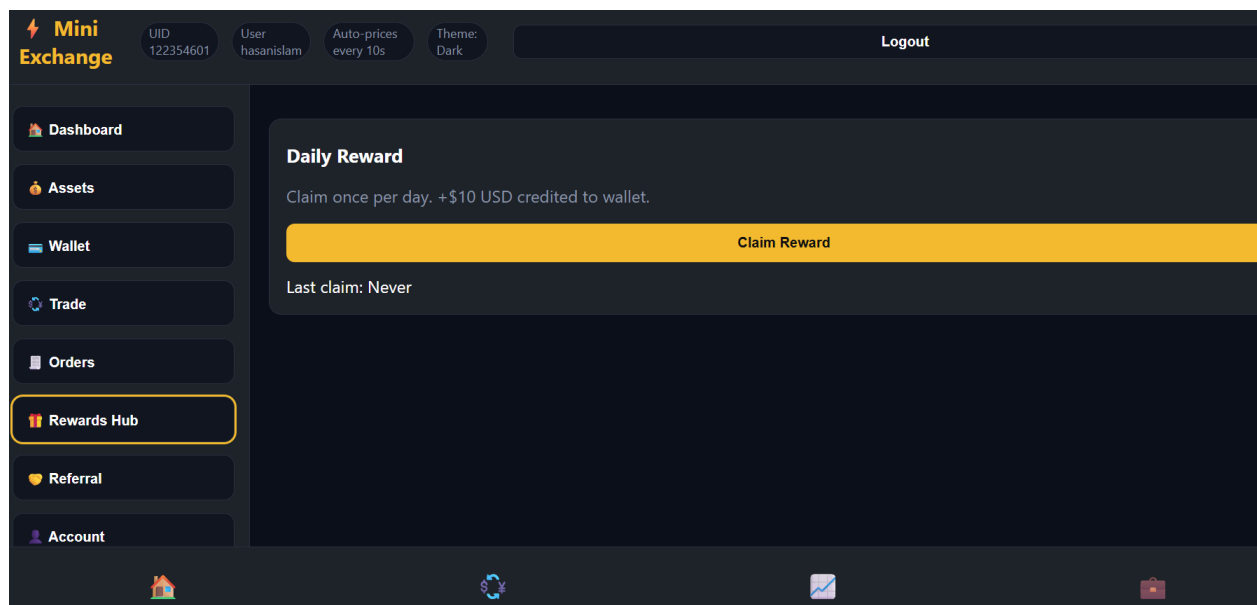
4. Trade: Access to Spot, Futures, and Margin trading tabs.



5. Orders: Keeps track of trades (placeholder).



6. Rewards Hub: Earn rewards up to 10% APY for deposits.



7. Referral: Invite and earn bonus.

⚡ Mini Exchange

UID: 122354601

User: hasanislam

Auto-prices every 10s

Theme: Dark

Logout

Dashboard

Assets

Wallet

Trade

Orders

Rewards Hub

Referral

Account

Your Referral

BWLV5K

Generate / Refresh

Bonus: +\$5 USD for each referred user you add below.

Add Referred User

Username

friend123

Add Referral

Referred Users

8. Account: User profile and settings.

⚡ Mini Exchange

UID: 122354601

User: hasanislam

Auto-prices every 10s

Theme: Dark

Logout

Dashboard

Assets

Wallet

Trade

Orders

Rewards Hub

Referral

Account

Profile

Username

hasanislam

Save

Security

Enable 2FA (simulated)

This is a demo toggle; no codes are required.

9.Sub-Accounts:

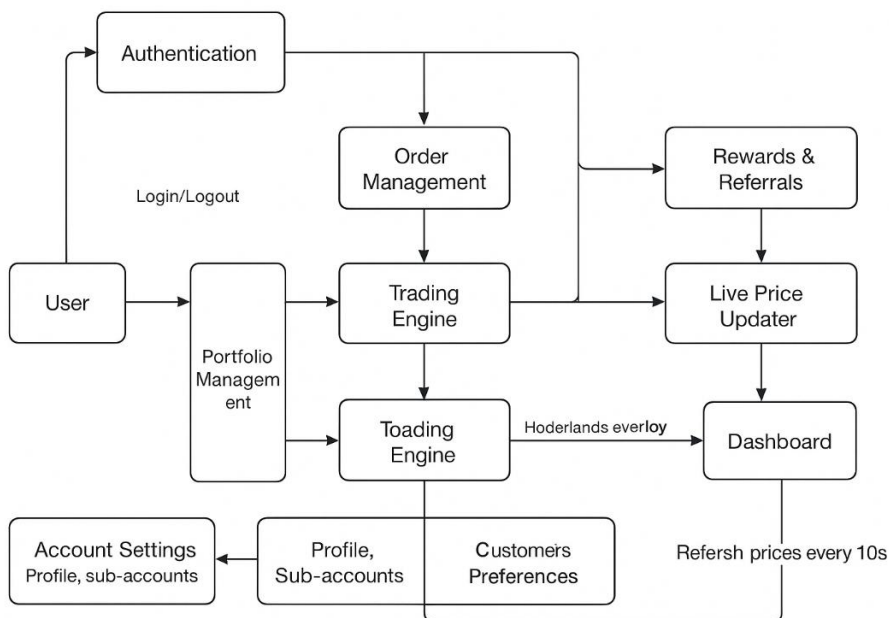
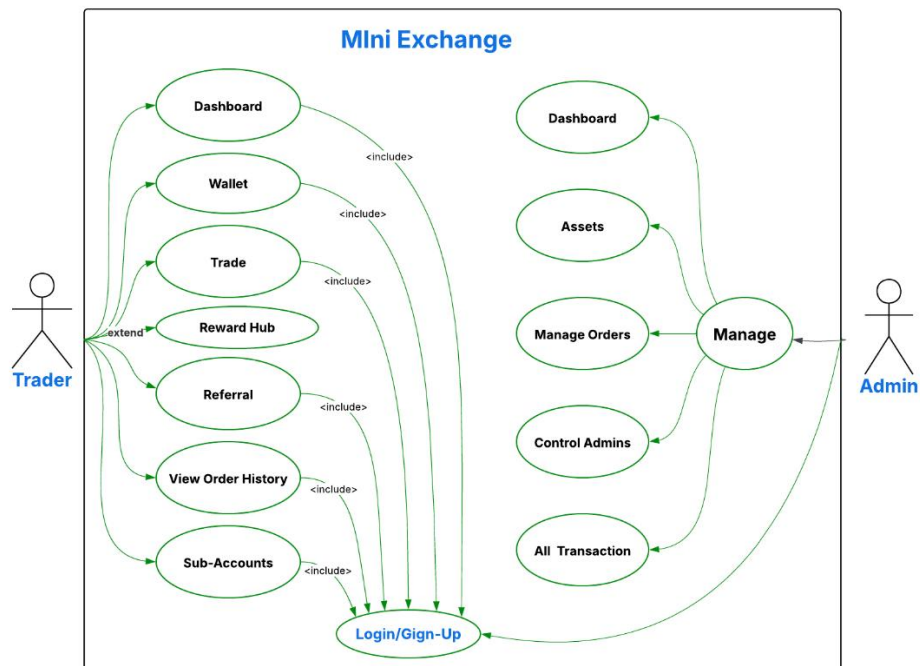
The screenshot displays the 'Sub-Accounts' section of a dashboard. On the left is a sidebar with navigation links: Dashboard, Assets, Wallet, Trade, Orders, Rewards Hub, Referral, Account, and Sub-Accounts (highlighted). The main content area is divided into three panels. The top-left panel, titled 'Create Sub-Account', contains a 'Sub-account name' input field with the text 'Bot-1' and a yellow 'Create' button. The top-right panel, titled 'Transfer Between Accounts', has fields for 'From' (Main), 'To' (Main), 'Asset' (USD), and 'Amount' (10), followed by a yellow 'Transfer' button. The bottom panel, titled 'All Accounts', is currently empty. A footer bar at the bottom contains four icons: a house, a circular arrow, a line graph, and a briefcase.

10.Settings:

The screenshot shows the 'Settings' section of the dashboard. The sidebar on the left is identical to the previous screenshot, with 'Sub-Accounts' highlighted. The main content area has two panels. The 'Theme' panel on the left features a yellow 'Toggle Dark / Light' button. The 'Data' panel on the right contains three buttons: 'Export JSON' (dark grey), 'Import JSON' (light grey), and 'Factory Reset' (red). The footer bar at the bottom is also identical to the previous screenshot.

- **Interactive Elements:**
 1. **Wallet Actions:** Deposit, Withdraw, Trade, Transfer buttons.
 2. **Tab Switching:** Spot / Futures / Margin with dynamic highlighting.
 3. **Auto-price refresh every 10 seconds**
 4. **Reward Card Section** with APY display.

4. Dataflow Diagram



5. Validation & Security

- **Validation Implemented:**

1. PHP backend validation ensures that all input fields such as registration, login, deposit, and withdrawal are correctly filled before processing.
2. JavaScript form validation prevents submission of empty or invalid fields in real time.
3. Backend and frontend validation work together to ensure consistency of user data.
4. Auto-price update and trade functions are verified through backend checks to avoid invalid requests.
5. All wallet actions (Deposit, Withdraw, Trade, Transfer) are validated against user balance and transaction limits.

- **Security Notes:**

1. Secure authentication implemented using PHP sessions and password hashing (password_hash() and password_verify())
2. All database queries use prepared statements to prevent SQL injection.
3. User sessions are protected from unauthorised access through session timeout and logout handling.
4. Sensitive data such as passwords and balances are never exposed to the client side.
5. Future enhancements: implementation of two-factor authentication (2FA) and encrypted API integration for real-time trading data.

6. Test Cases

The testing and evaluation phase for the Mini Crypto Exchange Web Application validated the design, functionality, and performance against the specified requirements. The following summarises the results:

Requirements Overview

- **System Requirements:**

The system successfully handled user authentication, wallet management, trading actions, and reward updates. The PHP and MySQL integration ensured secure and accurate data handling.

- **User Requirements:**

Users were able to view balances, trade, deposit, withdraw, and check rewards. Admins could view user data, manage assets, and control prices effectively.

- **Functional Requirements:**
Features like login/logout, deposit, withdraw, price updates, and database synchronisation worked as expected.
- **Non-Functional Requirements:**
The application ran smoothly on multiple browsers, ensuring responsiveness and dark-mode compatibility.
- **Feature Identification**
- **User Panel:**
Modules such as registration, login, wallet management, trade, and referral system were tested and worked properly.
- **Admin Panel:**
Functions including user monitoring, asset management, and reward updates were fully operational.
- **Work Plan**
- The project was developed in stages — planning, frontend design, backend integration, database setup, testing, and deployment. Regular testing ensured bugs were fixed early and all components interacted correctly.
- Budget
- The project was completed within budget, covering software, hosting, and server setup. No major cost changes occurred during development.
- **Development Model**
- The **Spiral Model** was used, involving iterative cycles of planning, coding, testing, and evaluation. This allowed improvements and added stability to the final application.

Test Case Summary

Test Case	Input	Expected Result	Actual Result
User Login	Enter valid username & password	Redirects to dashboard	Successful
Deposit	Enter amount \$100	Wallet increases by \$100	Successful
Withdraw	Enter amount \$50	Wallet decreases by \$50	Successful
Trade	Select "BTC to ETH"	Trade executed and updated in DB	Successful
Price Auto-Refresh	Wait 10 seconds	Price updates displayed	Successful
Admin Login	Enter admin credentials	Redirects to admin panel	Successful
Add Asset	New asset details	Asset saved and listed	Successful
Update Reward	Change APY value	Display updates in reward hub	Successful
Logout	Click "Logout"	Session destroyed, redirect to login	Successful

7. Reflection

Developing this Mini Crypto Exchange Web Application was a great experience and it enhanced my knowledge of full-stack development.

I made the frontend (HTML, CSS and JS) & backend integration working, with a MySQL DataBase. Developing an admin and user module enhanced my grip in role based access control and data security.

The most difficult parts were real-time updating, making a clean looking dark themed interface and optimising to deal with more than one user on the database.

Skills this project brought me It helped improve my database design, PHP programming and I'm sure I'll be able to extend it further by adding API integration (so real prices), more advanced user experience like an authentication system, etc.

(Note: I will only have the frontend side up on my Github, for the backend + frontend use the zip file)

<https://iftakharulislam909-eng.github.io/my-exchange-system/>

8. Reference

//Black Box// (project reference guidelines).

Online documentation for HTML, CSS JavaScript, PHP and My SQL